Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: W-D Ranch, Inc.

Physical address of dairy:

Number and Street	City	County	Zip Code
137 Texas RD	Modesto	Stanislaus	95358

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1948

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0012-0052-0007-0000

B. OPERATORS

David, Ulm

Operator name: David, Ulm

Telephone no.: (209) 523-0075 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
137 Texas RD	Modesto	CA	95358

This operator is responsible for paying permit fees.

C. OWNERS

David, Ulm

Legal owner name: David, Ulm

Telephone no.: (209) 523-0075 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
137 Texas RD	Modesto	CA	95358

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	50	75	0	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	50	75	0	0
Average number	0	0	25	50	0	0
Avg live weight (lbs)	0	0	1,100	900		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 775.80 *tons per reporting period*

Total nitrogen from manure: 7,117.50 *lbs per reporting period*

After ammonia losses (30% loss applied): 4,982.25 *lbs per reporting period*

Total phosphorus from manure: 1,204.50 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 3,715,000 *gallons*

Total nitrogen generated: 6,217.47 *lbs*

Total phosphorus generated: 2,327.74 *lbs*

Total potassium generated: 18,610.02 *lbs*

Total salt generated: 85,287.39 *lbs*

	3,715,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	3,715,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal Sample	Surface water
Dairy Well	Ground water
House Well	Ground water
Soaker Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	23	23	2	none	0012-0052-0006-0000
Field 2	30	30	2	process wastewater	0012-0052-0007-0000
Field 3	19	19	2	process wastewater	0012-0052-0007-0000
Field 4	35	35	2	process wastewater	0012-0052-0007-0000
Field 5	20	20	2	process wastewater	0012-0052-0007-0000
Totals for areas that were used for application	104	104	8		
Totals for areas that were not used for application	23	23	2		
Land application area totals	127	127	10		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

11/13/2022: Oats, hay

Crop: Oats, hay Acres planted: 23 Plant date: 11/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	79.25 *ton*	Dry-weight		10.0	13,700.00	4,600.00	30,900.00		13.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	4.00
Total actual harvest content	3.45	84.97	28.53	191.65	824.89

Field 1

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 23 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	701.54 *ton*	Dry-weight		67.7	9,000.00	2,600.00	13,800.00		5.38

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	28.00
Total actual harvest content	30.50	177.34	51.23	271.92	1,060.08

Field 2

Field name: Field 2

11/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	525.00 *ton*	Dry-weight		74.2	12,100.00	3,300.00	18,000.00		8.76

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	16.00
Total actual harvest content	17.50	109.26	29.80	162.54	791.03

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	967.97 *ton*	Dry-weight		70.1	7,400.00	3,500.00	19,900.00		7.33

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	30.00
Total actual harvest content	32.27	142.78	67.53	383.97	1,414.31

Field 3

Field name: Field 3

11/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 11/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	326.00 *ton*	Dry-weight		76.9	12,900.00	3,400.00	19,500.00		9.66

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	16.00
Total actual harvest content	17.16	102.26	26.95	154.58	765.74

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/02/2023	601.51 *ton*	Dry-weight		70.0	6,600.00	2,500.00	15,800.00		6.87

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	30.00
Total actual harvest content	31.66	125.37	47.49	300.12	1,304.96

Field 4

Field name: Field 4

11/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	626.85 *ton*	Dry-weight		75.4	14,100.00	3,900.00	27,900.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	16.00
Total actual harvest content	17.91	124.25	34.37	245.85	889.98

Field 4

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/02/2023	1,153.91 *ton*	Dry-weight		64.8	9,400.00	3,100.00	10,900.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	30.00
Total actual harvest content	32.97	218.17	71.95	252.99	1,322.97

Field 5

Field name: Field 5

11/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	356.30 *ton*	Dry-weight		76.2	16,300.00	4,000.00	27,700.00		9.69

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	16.00
Total actual harvest content	17.82	138.22	33.92	234.89	821.71

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/02/2023	557.02 *ton*	Dry-weight		69.7	7,600.00	3,300.00	14,100.00		6.45

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	30.00
Total actual harvest content	27.85	128.27	55.70	237.98	1,088.61

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 11/13/2022: Oats, hay

Field name: Field 1

Crop: Oats, hay — Plant date: 11/13/2022

No nutrient budget entered for this crop.

Field 1 - 06/10/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	2.07	0.00	0.00	43.40	5,695,882.97 *gal*
Application event totals		2.07	0.00	0.00	43.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	1.06	0.00	0.00	22.27	2,922,887.31 *gal*
Application event totals		1.06	0.00	0.00	22.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	1.06	0.00	0.00	22.27	2,922,887.31 *gal*
Application event totals		1.06	0.00	0.00	22.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	1.28	0.00	0.00	26.84	3,522,453.94 *gal*
Application event totals		1.28	0.00	0.00	26.84	

Field 1 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	1.28	0.00	0.00	26.84	3,522,453.94 *gal*
Application event totals		1.28	0.00	0.00	26.84	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	1.28	0.00	0.00	26.84	3,522,453.94 *gal*
Application event totals		1.28	0.00	0.00	26.84	

Field 2 - 11/13/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough Plant date: 11/13/2022

No nutrient budget entered for this crop.

Field 2 - 06/10/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	2.07	0.00	0.00	43.40	7,429,412.57 *gal*
Application event totals		2.07	0.00	0.00	43.40	

Field 2 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	15.55	8.19	92.34	354.79	249,600.00 *gal*
Canal Sample	Surface water	1.06	0.00	0.00	22.27	3,812,461.71 *gal*
Application event totals		16.61	8.19	92.34	377.06	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	19.90	6.00	31.57	186.52	374,600.00 *gal*
Canal Sample	Surface water	1.06	0.00	0.00	22.27	3,812,461.71 *gal*
Application event totals		20.96	6.00	31.57	208.79	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	1.99	0.60	3.16	18.67	37,500.00 *gal*
Canal Sample	Surface water	1.28	0.00	0.00	26.84	4,594,505.14 *gal*
Application event totals		3.27	0.60	3.16	45.51	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	1.28	0.00	0.00	26.84	4,594,505.14 *gal*
Application event totals		1.28	0.00	0.00	26.84	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	1.28	0.00	0.00	26.84	4,594,505.14 *gal*
Application event totals		1.28	0.00	0.00	26.84	

Field 3 - 11/13/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 11/13/2022

Field 3 - 11/13/2022: Oats, silage-soft dough

No nutrient budget entered for this crop.

Field 3 - 06/10/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	2.07	0.00	0.00	43.40	4,705,294.63 *gal*
Application event totals		2.07	0.00	0.00	43.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	3.01	1.59	17.87	68.68	30,600.00 *gal*
Canal Sample	Surface water	1.06	0.00	0.00	22.27	2,414,559.08 *gal*
Application event totals		4.07	1.59	17.87	90.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	21.31	6.43	33.80	199.69	254,000.00 *gal*
Canal Sample	Surface water	1.06	0.00	0.00	22.27	2,414,559.08 *gal*
Application event totals		22.37	6.43	33.80	221.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	3.52	1.06	5.59	33.02	42,000.00 *gal*
Canal Sample	Surface water	1.28	0.00	0.00	26.84	2,909,853.26 *gal*
Application event totals		4.80	1.06	5.59	59.86	

Field 3 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	15.78	4.76	25.03	147.88	188,100.00 *gal*
Canal Sample	Surface water	1.28	0.00	0.00	26.84	2,909,853.26 *gal*
Application event totals		17.06	4.76	25.03	174.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	1.28	0.00	0.00	26.84	2,909,853.26 *gal*
Application event totals		1.28	0.00	0.00	26.84	

Field 4 - 11/13/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough Plant date: 11/13/2022

No nutrient budget entered for this crop.

Field 4 - 06/10/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	2.07	0.00	0.00	43.40	8,667,648.00 *gal*
Application event totals		2.07	0.00	0.00	43.40	

Field 4 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	29.76	15.68	176.69	678.88	557,200.00 *gal*
Canal Sample	Surface water	1.06	0.00	0.00	22.27	4,447,872.00 *gal*
Application event totals		30.82	15.68	176.69	701.15	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	25.07	7.56	39.77	234.95	550,500.00 *gal*
Canal Sample	Surface water	1.06	0.00	0.00	22.27	4,447,872.00 *gal*
Application event totals		26.13	7.56	39.77	257.22	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	24.90	7.51	39.50	233.37	546,800.00 *gal*
Canal Sample	Surface water	1.28	0.00	0.00	26.84	5,360,256.00 *gal*
Application event totals		26.18	7.51	39.50	260.21	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	1.28	0.00	0.00	26.84	5,360,256.00 *gal*
Application event totals		1.28	0.00	0.00	26.84	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	1.28	0.00	0.00	26.84	5,360,256.00 *gal*
Application event totals		1.28	0.00	0.00	26.84	

Field 5 - 11/13/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough

Plant date: 11/13/2022

Field 5 - 11/13/2022: Oats, silage-soft dough

No nutrient budget entered for this crop.

Field 5 - 06/10/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	2.07	0.00	0.00	43.40	4,952,941.71 *gal*
Application event totals		2.07	0.00	0.00	43.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	22.24	11.72	132.08	507.45	238,000.00 *gal*
Canal Sample	Surface water	1.06	0.00	0.00	22.27	2,541,641.14 *gal*
Application event totals		23.30	11.72	132.08	529.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	32.44	9.78	51.47	304.05	407,100.00 *gal*
Canal Sample	Surface water	1.06	0.00	0.00	22.27	2,541,641.14 *gal*
Application event totals		33.50	9.78	51.47	326.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	19.05	5.74	30.22	178.50	239,000.00 *gal*
Canal Sample	Surface water	1.28	0.00	0.00	26.84	3,063,003.43 *gal*
Application event totals		20.33	5.74	30.22	205.34	

Field 5 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	1.28	0.00	0.00	26.84	3,063,003.43 *gal*
Application event totals		1.28	0.00	0.00	26.84	

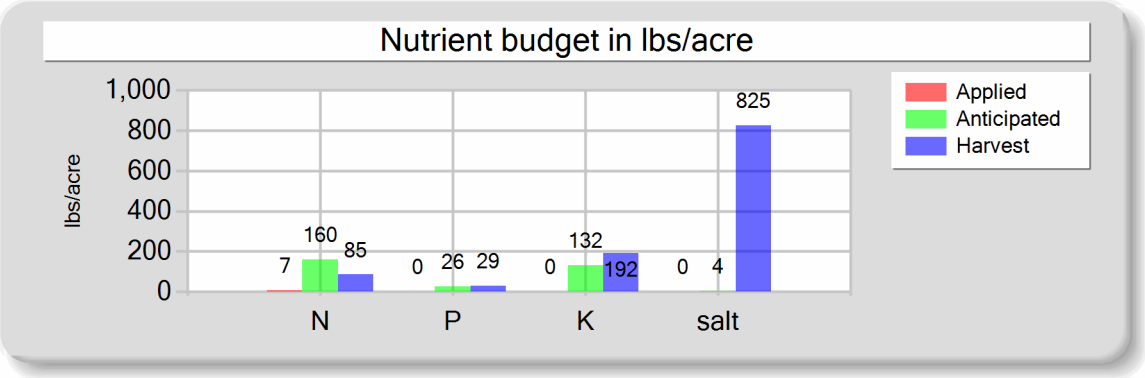
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	1.28	0.00	0.00	26.84	3,063,003.43 *gal*
Application event totals		1.28	0.00	0.00	26.84	

B. NUTRIENT BUDGET

Field 1 - 11/13/2022: Oats, hay

Field name: Field 1 Crop: Oats, hay Plant date: 11/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	7.00	0.00	0.00	0.00
Anticipated crop nutrient removal	160.00	26.00	132.00	4.00
Actual crop nutrient removal	84.97	28.53	191.65	824.89
Nutrient balance	-77.97	-28.53	-191.65	-824.89
Applied to removed ratio	0.08	0.00	0.00	0.00

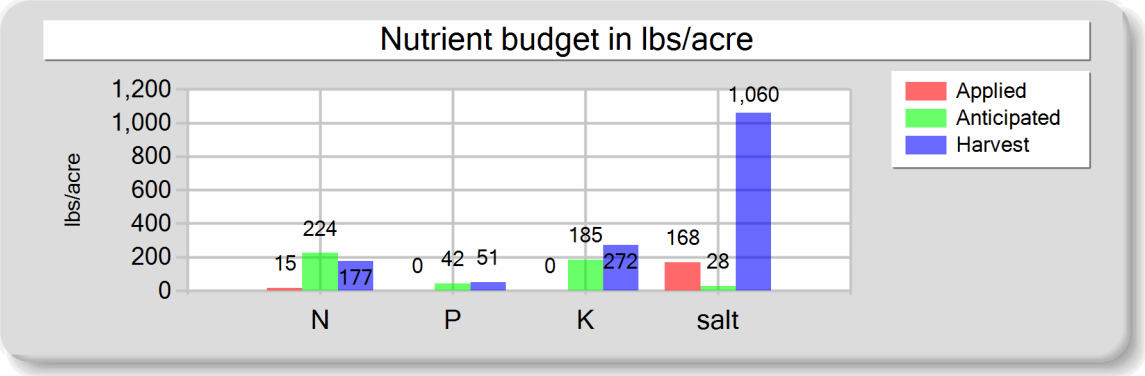
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 1 - 06/10/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.02	0.00	0.00	168.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	15.02	0.00	0.00	168.46
Anticipated crop nutrient removal	224.00	42.00	184.80	28.00
Actual crop nutrient removal	177.34	51.23	271.92	1,060.08
Nutrient balance	-162.32	-51.23	-271.92	-891.63
Applied to removed ratio	0.08	0.00	0.00	0.16

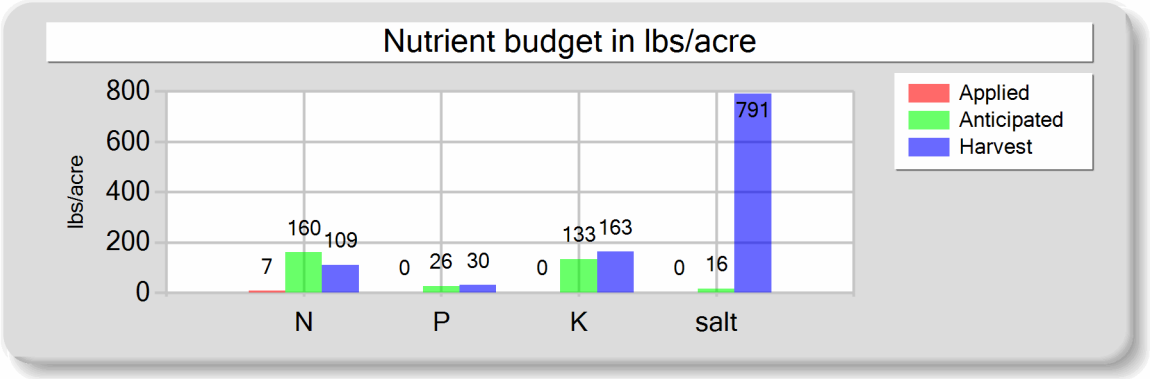
Fresh water applied	
22,109,019.41	*gallons*
814.20	*acre-inches*
35.40	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/13/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 11/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	7.00	0.00	0.00	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	16.00
Actual crop nutrient removal	109.26	29.80	162.54	791.03
Nutrient balance	-102.26	-29.80	-162.54	-791.03
Applied to removed ratio	0.06	0.00	0.00	0.00

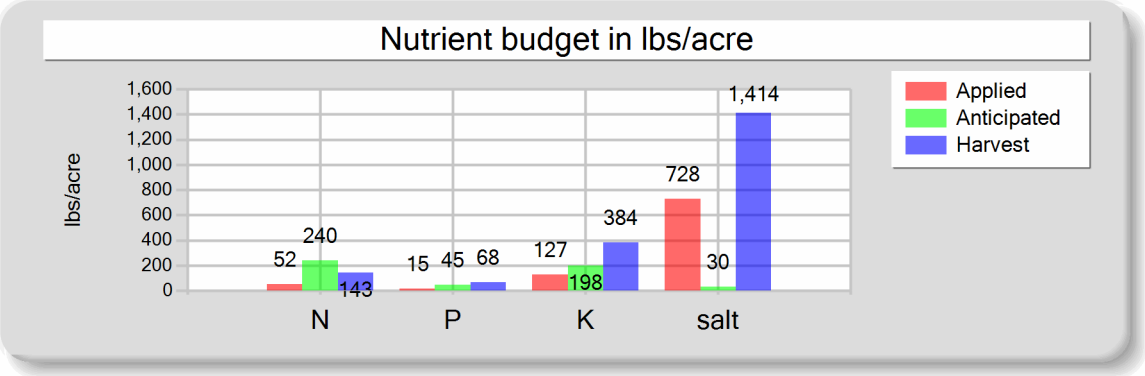
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 2 - 06/10/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	37.45	14.80	127.08	559.98
Fresh water	8.02	0.00	0.00	168.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	52.47	14.80	127.08	728.44
Anticipated crop nutrient removal	240.00	45.00	198.00	30.00
Actual crop nutrient removal	142.78	67.53	383.97	1,414.31
Nutrient balance	-90.31	-52.74	-256.89	-685.88
Applied to removed ratio	0.37	0.22	0.33	0.52

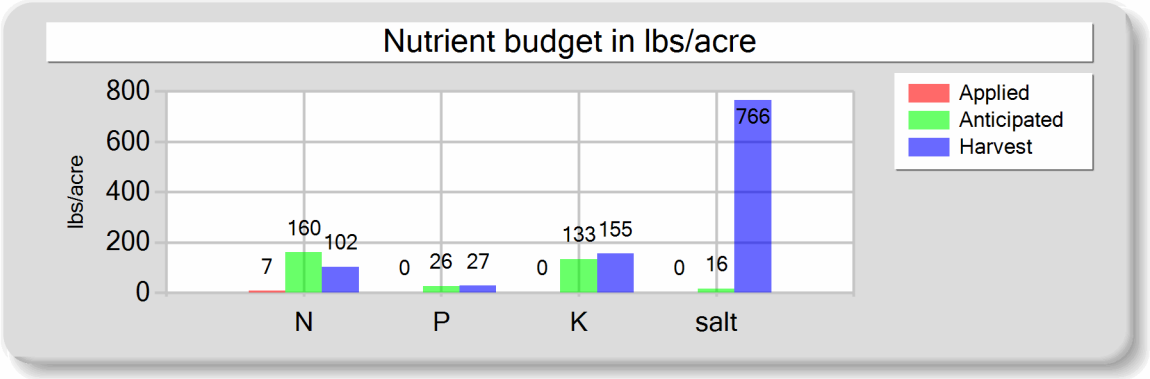
Fresh water applied	
28,837,851.41	*gallons*
1,062.00	*acre-inches*
35.40	*inches/acre*

Process wastewater applied	
661,700.00	*gallons*
24.37	*acre-inches*
0.81	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/13/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 11/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	7.00	0.00	0.00	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	16.00
Actual crop nutrient removal	102.26	26.95	154.58	765.74
Nutrient balance	-95.26	-26.95	-154.58	-765.74
Applied to removed ratio	0.07	0.00	0.00	0.00

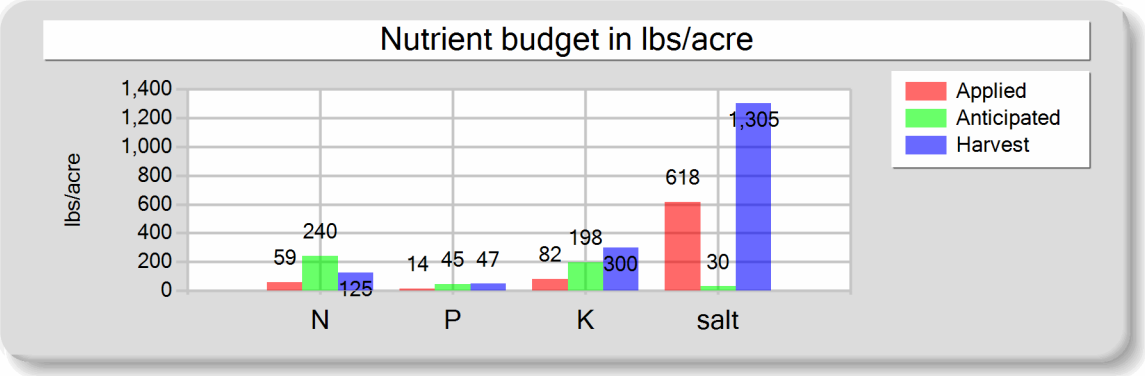
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 3 - 06/10/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	43.62	13.83	82.30	449.27
Fresh water	8.02	0.00	0.00	168.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	58.64	13.83	82.30	617.73
Anticipated crop nutrient removal	240.00	45.00	198.00	30.00
Actual crop nutrient removal	125.37	47.49	300.12	1,304.96
Nutrient balance	-66.72	-33.65	-217.82	-687.23
Applied to removed ratio	0.47	0.29	0.27	0.47

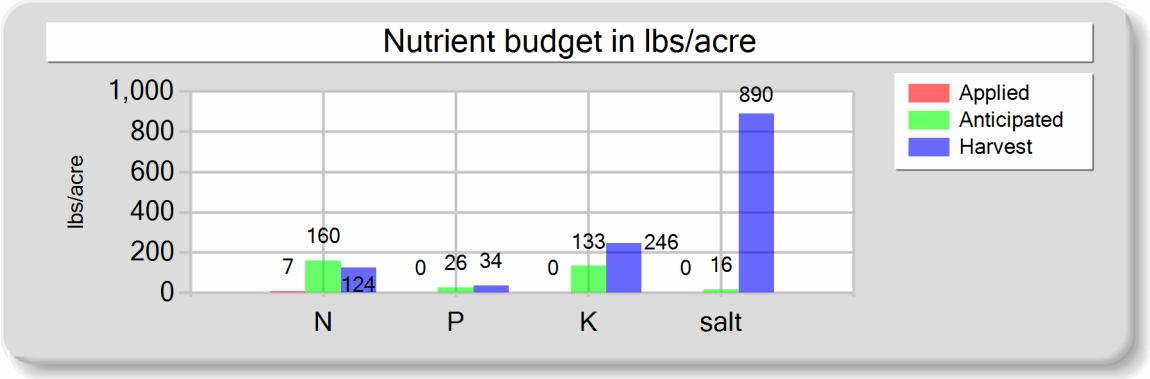
Fresh water applied	
18,263,972.57	*gallons*
672.60	*acre-inches*
35.40	*inches/acre*

Process wastewater applied	
514,700.00	*gallons*
18.95	*acre-inches*
1.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/13/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 11/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	7.00	0.00	0.00	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	16.00
Actual crop nutrient removal	124.25	34.37	245.85	889.98
Nutrient balance	-117.25	-34.37	-245.85	-889.98
Applied to removed ratio	0.06	0.00	0.00	0.00

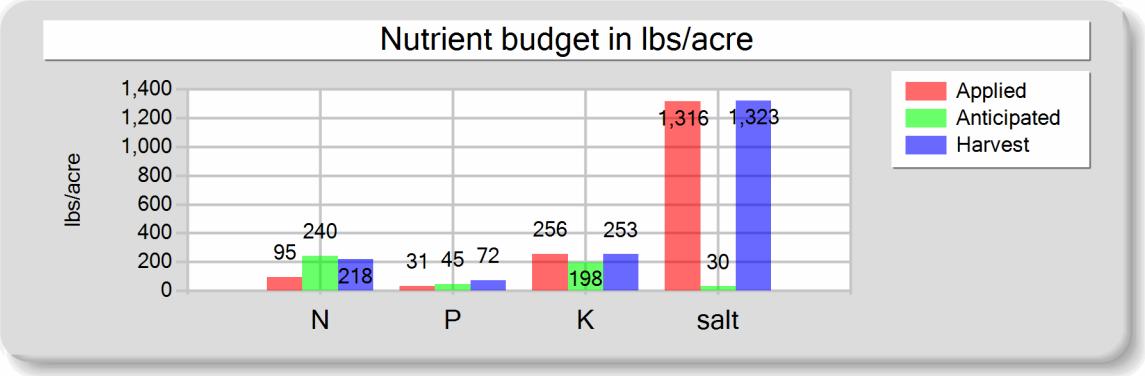
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 4 - 06/10/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	79.73	30.75	255.97	1,147.19
Fresh water	8.02	0.00	0.00	168.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	94.75	30.75	255.97	1,315.65
Anticipated crop nutrient removal	240.00	45.00	198.00	30.00
Actual crop nutrient removal	218.17	71.95	252.99	1,322.97
Nutrient balance	-123.42	-41.20	2.98	-7.33
Applied to removed ratio	0.43	0.43	1.01	0.99

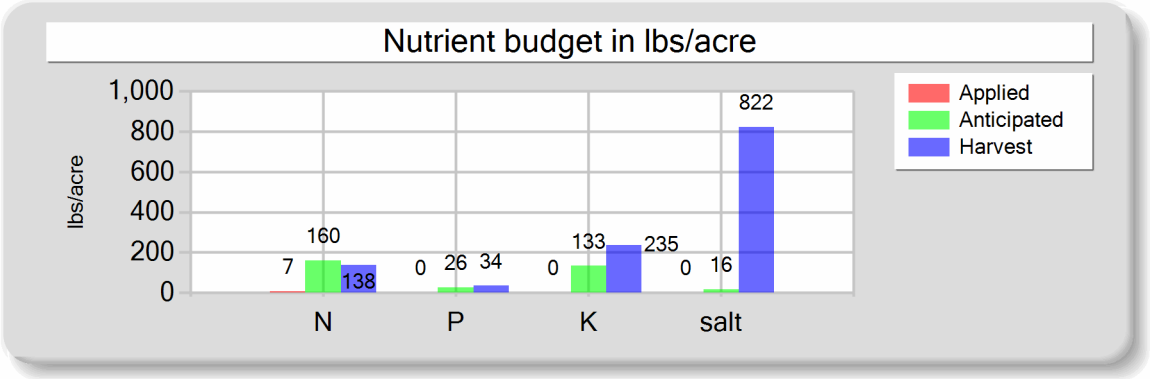
Fresh water applied	
33,644,160.00	*gallons*
1,239.00	*acre-inches*
35.40	*inches/acre*

Process wastewater applied	
1,654,500.00	*gallons*
60.93	*acre-inches*
1.74	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 11/13/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 11/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	7.00	0.00	0.00	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	16.00
Actual crop nutrient removal	138.22	33.92	234.89	821.71
Nutrient balance	-131.22	-33.92	-234.89	-821.71
Applied to removed ratio	0.05	0.00	0.00	0.00

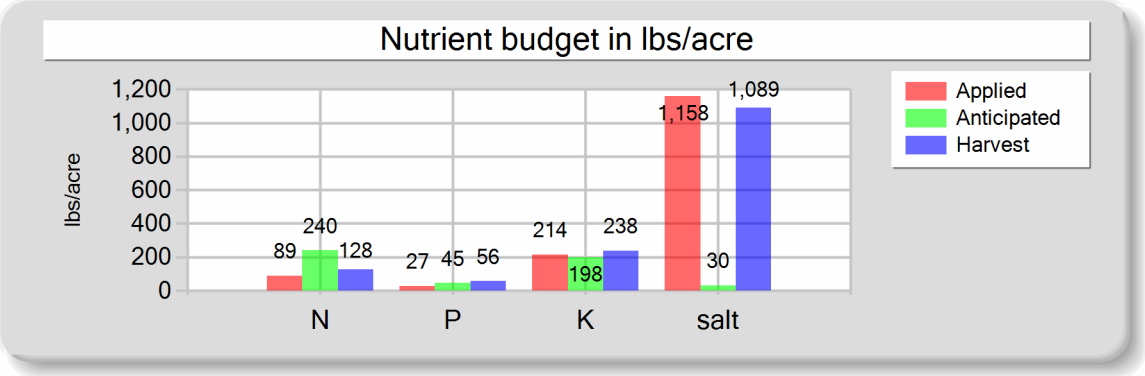
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 5 - 06/10/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	73.74	27.25	213.76	990.01
Fresh water	8.02	0.00	0.00	168.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	88.76	27.25	213.76	1,158.46
Anticipated crop nutrient removal	240.00	45.00	198.00	30.00
Actual crop nutrient removal	128.27	55.70	237.98	1,088.61
Nutrient balance	-39.51	-28.45	-24.22	69.85
Applied to removed ratio	0.69	0.49	0.90	1.06

Fresh water applied	
19,225,234.28	*gallons*
708.00	*acre-inches*
35.40	*inches/acre*

Process wastewater applied	
884,100.00	*gallons*
32.56	*acre-inches*
1.63	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

No manure analyses entered.

B. PROCESS WASTEWATER ANALYSES

1st Quarter Lagoon

Sample and source description: 1st Quarter Lagoon

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	416.00	93.40	0.00	2.00	77.40	992.00								6,190.00	4,110
DL	10.00	2.00	0.01	2.00	0.20	0.50								0.10	10

2nd Quarter Lagoon

Sample and source description: 2nd Quarter Lagoon

Sample date: 06/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	222.00	98.30	0.00	2.00	118.00	1,330.00								7,690.00	5,110
DL	10.00	2.00	0.01	2.00	0.20	0.50								0.10	10

3rd Quarter Lagoon

Sample and source description: 3rd Quarter Lagoon

Sample date: 09/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	189.00	186.00	0.00	2.00	57.60	303.00								2,690.00	1,790
DL	10.00	2.00	0.01	2.00	0.20	0.50								0.10	10

4th Quarter Lagoon

Sample and source description: 4th Quarter Lagoon

Sample date: 11/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	164.00	89.10	0.00	2.00	61.40	653.00								3,890.00	2,580
DL	10.00	2.00	0.01	2.00	0.20	0.50								0.10	10

C. FRESH WATER ANALYSES

Canal Sample

Canal Sample

Sample description: Canal Sample

Sample date: 09/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value												21
DL												1

MID Lateral 4

Sample description: MID Lateral 4

Sample date: 09/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			0.10	1.70	1.10	0.10	13.70		0.80	2.40	30.00	21
DL			0.01	0.01	0.01	0.01	0.01		0.01	0.01	0.01	1

Dairy Well

Dairy Well

Domestic. Well

Sample description: Domestic. Well

Sample date: 09/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			4.30	31.00	9.20	33.60	157.00		21.40	13.80	400.00	275
DL			0.01	0.01	0.01	0.01	0.01		0.01	0.01	0.01	1

House Well

Domestic Well

Sample description: Domestic Well

Sample date: 09/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			7.70	48.60	17.80	47.60	260.00		26.40	17.50	620.00	427
DL			0.01	0.01	0.01	0.01	0.01		0.01	0.01	0.01	1

Soaker Well

Domestic Well

Sample description: Domestic Well

Sample date: 09/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			29.80	105.00	39.80	91.10	495.00		37.10	48.90	1,230.00	846
DL			0.01	0.01	0.01	0.01	0.01		0.01	0.01	0.01	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 1 - 11/13/2022: Oats, hay

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	4,600.00	30,900.00		13.30
DL	0.05	0.02	0.02		0.05

Field 1 - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,000.00	2,600.00	13,800.00		5.38
DL	0.05	0.02	0.05		0.05

Field 2 - 11/13/2022: Oats, silage-soft dough

Field 2 - 11/13/2022: Oats, silage-soft dough

Winter Forage Hay

Sample and source description: Winter Forage Hay

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	3,300.00	18,000.00		8.76
DL	0.05	0.02	0.02		0.05

Field 2 - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,400.00	3,500.00	19,900.00		7.33
DL	0.05	0.02	0.05		0.05

Field 3 - 11/13/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	3,400.00	19,500.00		9.66
DL	0.05	0.02	0.05		0.05

Field 3 - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,600.00	2,500.00	15,800.00		6.87
DL	0.05	0.02	0.05		0.05

Field 4 - 11/13/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,100.00	3,900.00	27,900.00		10.10
DL	0.05	0.02	0.02		0.05

Field 4 - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,400.00	3,100.00	10,900.00		5.70
DL	0.05	0.02	0.05		0.05

Field 5 - 11/13/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,300.00	4,000.00	27,700.00		9.69
DL	0.05	0.02	0.05		0.05

Field 5 - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,600.00	3,300.00	14,100.00		6.45
DL	0.05	0.02	0.05		0.05

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

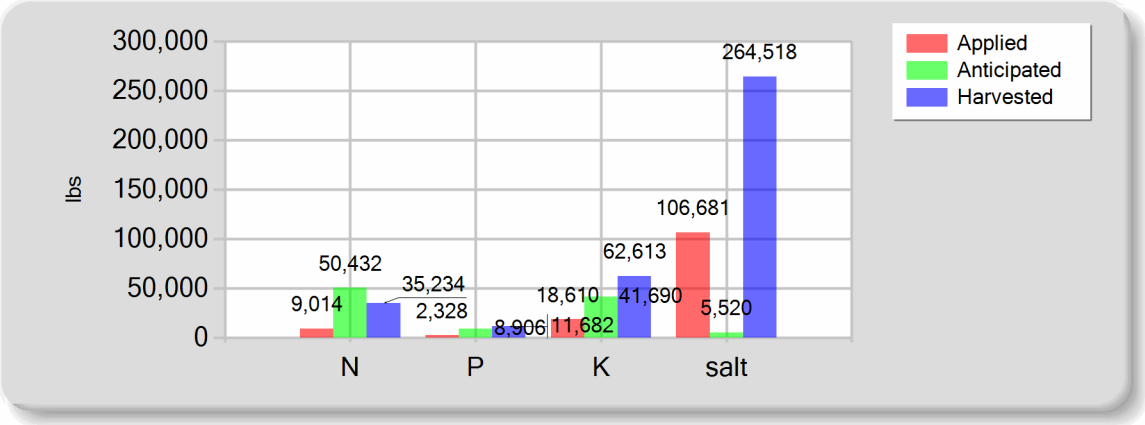
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

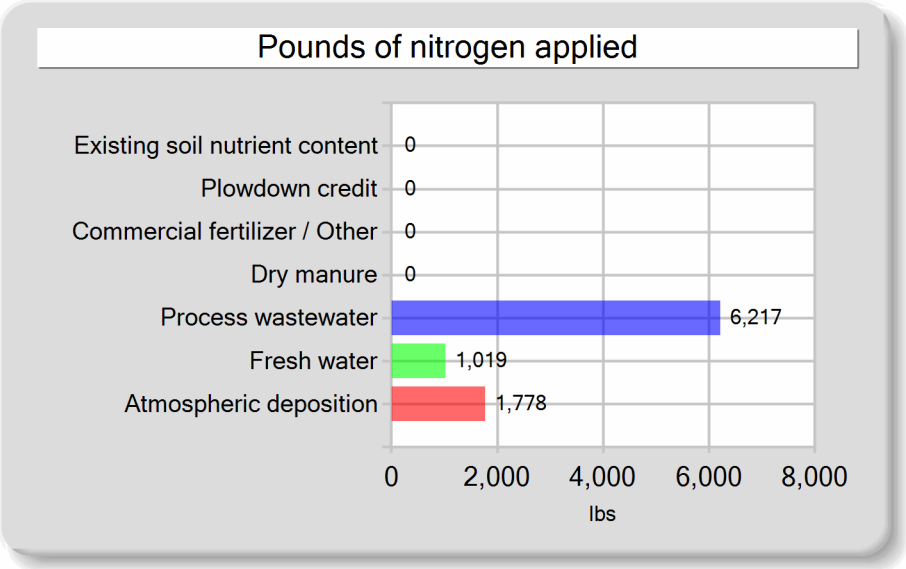
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

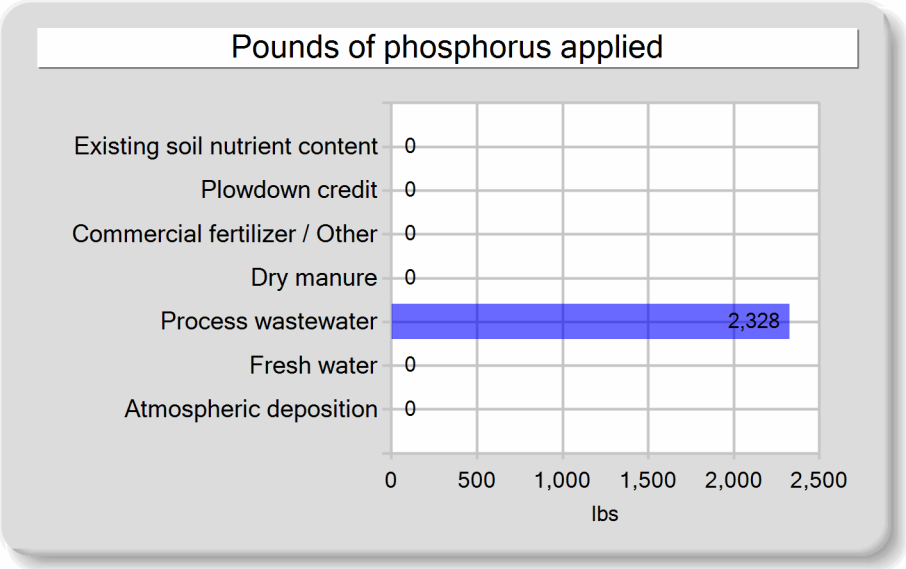
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	6,217.47	2,327.74	18,610.02	85,287.39
Fresh water	1,018.76	0.00	0.00	21,393.95
Atmospheric deposition	1,778.00	0.00	0.00	0.00
Total nutrients applied	9,014.23	2,327.74	18,610.02	106,681.34
Anticipated crop nutrient removal	50,432.00	8,906.40	41,689.60	5,520.00
Actual crop nutrient removal	35,233.85	11,682.19	62,613.15	264,517.85
Nutrient balance	-26,219.63	-9,354.45	-44,003.12	-157,836.51
Applied to removed ratio	0.26	0.20	0.30	0.40

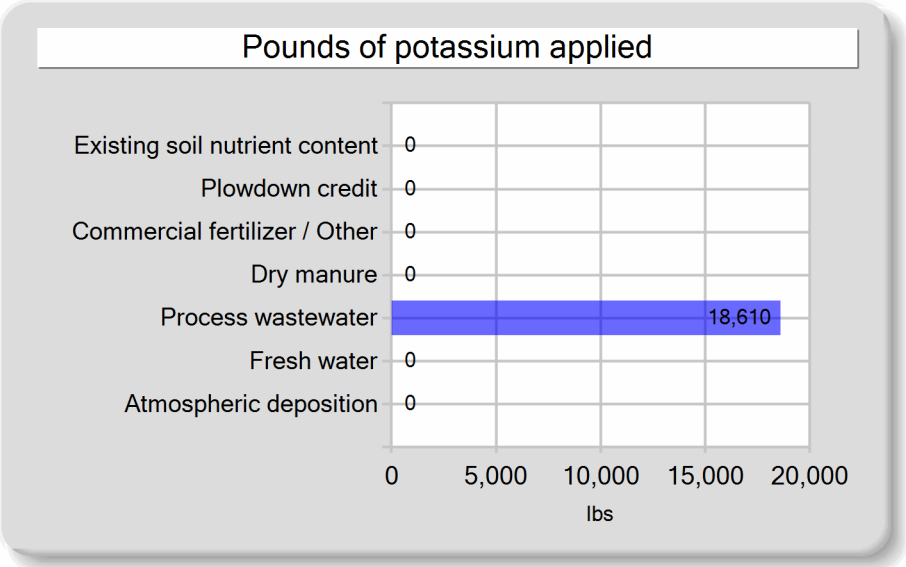
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

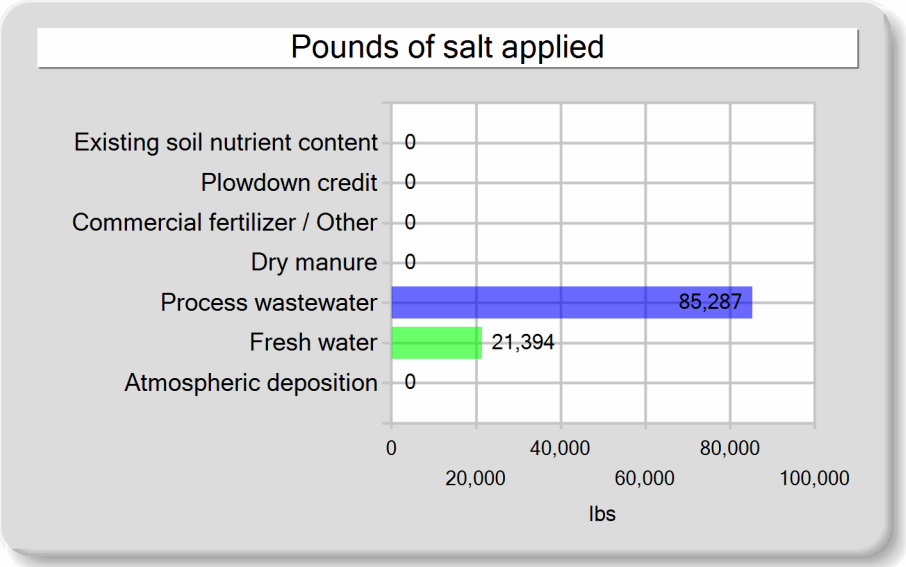


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Milk Cows and Dry cows were sold August 17, 2022

Remaining Heifers were sold November 9, 2023

Little solid manure was generated, no solid manure was sampled and no solid manure was applied or exported.

Report has Validation errors because there are no applications of solid manure , liquid manure or fresh water on the winter crops.

ANNUAL REPORT VALIDATION INFORMATION

A. VALIDATION ERRORS

The following sections contain validation errors and should be reviewed before submitting the Annual Report :

1. Land Application Events

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Ulm David	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
DATE	DATE

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

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SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Ulm David	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
10-27-24	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

A & L Western Laboratories, Inc.

1311 Woodland Avenue, Modesto CA 95351 209-529-4080
10220 SW Nimbus Avenue Bldg. K-9, Portland OR 97223 503-968-9225

Ag Suitability Water Analysis

W-D RANCH
137 TEXAS RD
MODESTO, CA 95358

Order Number: 67145
Lab Number: W67145-01
Submitted Date: 9/28/2023
Report Date: 10/6/2023
Submitted By: 597

Grower: W-D RANCH

Description: DAIRY WELL

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

Analytes				Normal Values	Problem Values
pH (pH)		7.5	pH units	6.5 to 8.0	<6.5 or >8.0
Electrical Conductivity (EC)		0.40	dS/m	0.5 to 3.0	<0.5 or >3.0
Boron (B)		0.06	mg/L	<0.5	>0.5
Cations	**mg/L**	**meq/L**	**lbs/ac-ft**	**Normal Values mg/L**	**Problem Values mg/L**
Calcium (Ca)	31.0	1.55	83.7	30 to 400	<30
Magnesium (Mg)	9.2	0.77	24.9	1 to 60	[Mg]>[Ca]
Sodium (Na)	33.6	1.46	90.7	<70	>70
Potassium (K)	3.3	0.08	8.8		
Anions	**mg/L**	**meq/L**	**lbs/ac-ft**	**Normal Values mg/L**	**Problem Values mg/L**
Bicarbonate (HCO_3)	157	2.57	423	<150	>300
Chloride (Cl)	13.8	0.39	37.1	<150	>200
Nitrate - Nitrogen (NO_3-N)	4.3	0.31	11.6	<10	>10
Sulfate - Sulfur (SO_4-S)	21.4	0.67	57.8		
Calculated Values				**Normal Values**	**Problem Values**
Total Dissolved Solids (TDS)		275	mg/L	1 to 1,500	>1,900
Sodium Absorption Ratio (SAR)		1.4		<6.0	>6.0
SAR/EC Ratio (SEC)		3.4		<5.0	>10.0
pHc (pHc)		7.8		<8.4 may add Ca	>8.4 may remove Ca

Gypsum Requirements	100% gypsum equivalent (lbs/ac-ft)
Eatons Gypsum Requirement (EGR)	96
Residual Sodium Carbonate (RSC)	59

RED = Value of Concern

contact@vtaglab.com

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10220 SW Nimbus Avenue Bldg. K-9, Portland OR 97223 503-968-9225

Ag Suitability Water Analysis

W-D RANCH
137 TEXAS RD
MODESTO, CA 95358

Order Number: 67145
Lab Number: W67145-02
Submitted Date: 9/28/2023
Report Date: 10/6/2023
Submitted By: 597

Grower: W-D RANCH

Description: HOUSE WELL

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

Analytes				Normal Values	Problem Values
pH (pH)		7.5	pH units	6.5 to 8.0	<6.5 or >8.0
Electrical Conductivity (EC)		0.62	dS/m	0.5 to 3.0	<0.5 or >3.0
Boron (B)		0.11	mg/L	<0.5	>0.5
Cations	**mg/L**	**meq/L**	**lbs/ac-ft**	**Normal Values mg/L**	**Problem Values mg/L**
Calcium (Ca)	48.6	2.43	131	30 to 400	<30
Magnesium (Mg)	17.8	1.48	48.0	1 to 60	[Mg]>[Ca]
Sodium (Na)	47.6	2.07	129	<70	>70
Potassium (K)	3.9	0.10	10.5		
Anions	**mg/L**	**meq/L**	**lbs/ac-ft**	**Normal Values mg/L**	**Problem Values mg/L**
Bicarbonate (HCO_3)	260	4.27	703	<150	>300
Chloride (Cl)	17.5	0.49	47.3	<150	>200
Nitrate - Nitrogen (NO_3-N)	7.7	0.55	20.7	<10	>10
Sulfate - Sulfur (SO_4-S)	26.4	0.83	71.4		
Calculated Values				**Normal Values**	**Problem Values**
Total Dissolved Solids (TDS)		427	mg/L	1 to 1,500	>1,900
Sodium Absorption Ratio (SAR)		1.5		<6.0	>6.0
SAR/EC Ratio (SEC)		2.4		<5.0	>10.0
pHc (pHc)		7.3		<8.4 may add Ca	>8.4 may remove Ca

Gypsum Requirements	100% gypsum equivalent (lbs/ac-ft)
Eatons Gypsum Requirement (EGR)	63
Residual Sodium Carbonate (RSC)	84

RED = Value of Concern

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A & L Western Laboratories, Inc.

1311 Woodland Avenue, Modesto CA 95351 209-529-4080
10220 SW Nimbus Avenue Bldg. K-9, Portland OR 97223 503-968-9225

Ag Suitability Water Analysis

W-D RANCH
137 TEXAS RD
MODESTO, CA 95358

Order Number: 67145
Lab Number: W67145-03
Submitted Date: 9/28/2023
Report Date: 10/6/2023
Submitted By: 597

Grower: W-D RANCH

Description: SOAKER WELL

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

Analytes				Normal Values	Problem Values
pH (pH)		7.2	pH units	6.5 to 8.0	<6.5 or >8.0
Electrical Conductivity (EC)		1.23	dS/m	0.5 to 3.0	<0.5 or >3.0
Boron (B)		0.20	mg/L	<0.5	>0.5
Cations	**mg/L**	**meq/L**	**lbs/ac-ft**	**Normal Values mg/L**	**Problem Values mg/L**
Calcium (Ca)	105	5.27	285	30 to 400	<30
Magnesium (Mg)	39.8	3.32	108	1 to 60	[Mg]>[Ca]
Sodium (Na)	91.1	3.96	246	<70	>70
Potassium (K)	7.8	0.20	21.2		
Anions	**mg/L**	**meq/L**	**lbs/ac-ft**	**Normal Values mg/L**	**Problem Values mg/L**
Bicarbonate (HCO_3)	495	8.12	1340	<150	>300
Chloride (Cl)	48.9	1.38	132	<150	>200
Nitrate - Nitrogen (NO_3-N)	29.8	2.13	80.5	<10	>10
Sulfate - Sulfur (SO_4-S)	37.1	1.16	100		
Calculated Values				**Normal Values**	**Problem Values**
Total Dissolved Solids (TDS)		846	mg/L	1 to 1,500	>1,900
Sodium Absorption Ratio (SAR)		1.9		<6.0	>6.0
SAR/EC Ratio (SEC)		1.6		<5.0	>10.0
pHc (pHc)		6.8		<8.4 may add Ca	>8.4 may remove Ca

Gypsum Requirements	100% gypsum equivalent (lbs/ac-ft)
Eatons Gypsum Requirement (EGR)	0
Residual Sodium Carbonate (RSC)	0

RED = Value of Concern

contact@vtaglab.com

A & L WESTERN LABORATORIES, INC.

1311 Woodland Ave. #1 • Modesto, California 95351 • Phone 209-529-4080

CHAIN OF CUSTODY

Client: W-D Ranch
Address: 137 Texas Rd
Modesto Zip 95358

Phone: 209-404-0242
CC to: UhMSmailBox@gmail.com

Signature of person authorizing work under terms stated below*

*Net 30 days. All accounts past due will be subject to interest charges of 1.5% per month.
*Hazardous materials are the property of the client. The client is responsible for proper disposal of hazardous wastes. Clients not picking up hazardous wastes may be assessed an appropriate fee.

PROJECT ID

SAMPLED BY: (signature): David Ulm

Date	Time	sample type Grab	sample type Comp.	Sample Location	No. of Containers	ANALYSIS: W-1								REMARKS
9-27	10:30 AM			Dairy well		✓								
9-27	11:00 AM			House well		✓								
9-27	11:40 AM			Soaker well		✓								

Relinquished by (signature)	Date	Time	Received by (signature)	Date	Time
Relinquished by (signature): David Ulm	Date 9/28	Time 10:57AM	Received by (signature): Mar[illegible]	Date 9/28/25	Time 10:55
Relinquished by (signature):	Date	Time	Received by (signature):	Date	Time
Relinquished by (signature):	Date	Time	Received by (signature):	Date	Time
Relinquished by (signature):	Date	Time	Received by (signature):	Date	Time

Site Time: Start: Finish: Total Hours:

Driving Time: Start: Finish: Total Hours:

Dedicated Exclusively to Providing Quality Analytical Services